

CIS3032 - E-commerce and Digital Marketing

Research Article

The Impact of Product Pricing Strategies on Online Purchase Decisions Using E-Commerce Data



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Abstract

The proliferation of e-commerce platforms has intensified price competition and reshaped consumer purchase decision-making across two distinct but interdependent dimensions: the pre-purchase digital browsing environment and the post-commitment transactional outcome. This study investigates how product pricing and web-resident value indicators jointly influence online purchase decisions through a two-phase analytical framework. Phase 1 leverages the UCI Online Shoppers Purchasing Intention Dataset (12,330 user sessions) to model how session-level value indicators, principally PageValues and ProductRelated_Duration, predict purchase conversion (Revenue = True) using Logistic Regression and Random Forest classification algorithms. Phase 2 employs the UCI Online Retail Dataset (541,909 invoice line items; 4,372 unique customers; December 2010 to December 2011) to conduct macro-level transactional analysis, computing Price Elasticity of Demand via log-log linear regression and performing RFM-based K-Means customer segmentation.

Phase 1 findings reveal that PageValues is the strongest predictor of session conversion (Random Forest feature importance: 37.9%), with the classification model achieving an AUC-ROC of 0.92 and an overall accuracy of 0.90. Phase 2 findings confirm that the overall Price Elasticity of Demand is -0.52 (inelastic), with all three customer segments exhibiting near-uniform inelastic behaviour: VIP Wholesalers (Cluster A: $e = -0.50$), Frequent Mainstream Retailers (Cluster B: $e = -0.52$), and Low-Engagement Occasional Buyers (Cluster C: $e = -0.53$). Revenue concentration follows a Pareto pattern, with the top 8% of customers generating 62.3% of total platform revenue. The study contributes an integrated pricing-value framework bridging clickstream analytics and transactional economics, with direct implications for e-commerce pricing strategy, platform design, and customer retention investment.

Keywords: e-commerce pricing, value indicators, PageValues, price elasticity of demand, customer segmentation, RFM modelling, purchase conversion, K-Means clustering, online retail analytics

1. Introduction

The rapid expansion of e-commerce has revolutionised the retail sector, enabling firms to distribute goods and services across digital platforms while affording consumers unprecedented convenience, choice, and transactional efficiency. As online commerce has matured, competitive pressures have intensified significantly, compelling firms to adopt deliberate, data-informed strategies to attract customers, drive conversions, and sustain revenue growth. Among the many variables that shape a consumer's decision to purchase, price has historically been treated as the dominant lever. However, the digital environment introduces an additional and underexplored dimension: the financial value embedded within the browsing experience itself, prior to the moment of committed transaction.

Price tactics such as discount pricing, promotional pricing, dynamic pricing, and bundle pricing remain powerful mechanisms through which e-commerce firms modulate consumer demand. Yet in digital environments, the consumer's journey to the point of purchase is mediated by a layered web experience, in which the quality, relevance, and financial value signalled by the pages a customer encounters during a session exert measurable influence on their decision to convert. The concept of page-level value indicators, operationalised in the literature as metrics such as PageValues, represents a critical but understudied bridge between platform design and pricing strategy.

Prior research has addressed purchase prediction, customer segmentation, demand forecasting, and price optimisation largely as independent analytical problems. Tokuc and Dag [1] demonstrated that clickstream behavioural features reliably predict purchase intent; Lee et al. [2] confirmed that machine learning classifiers can identify conversion-driving session characteristics; and Wang and Zhang [3] established that user activity features outperform demographic indicators as purchase predictors. On the transactional side, Ozcan [4], John et al. [5], and Awaliyah et al. [6] demonstrated the power of RFM-based segmentation for identifying commercially distinct customer cohorts. However, none of these studies established a unified framework connecting front-end session value indicators with back-end pricing dynamics across the full e-commerce pipeline.

This study addresses that gap through a dual-phase design. Phase 1 models how value indicators embedded in web session data determine the probability of session conversion. Phase 2 quantifies how unit pricing shapes the volume of completed transactions across distinct customer segments identified through RFM clustering. The research objective is to examine how product pricing and web-resident value indicators jointly influence online purchase decisions, using an integrated two-dataset analytical framework that connects session-level conversion behaviour with transactional price sensitivity.

2. Background and Literature Review

2.1 Product Pricing Strategies in E-Commerce

Pricing is broadly recognised as one of the most influential determinants of consumer purchasing behaviour in digital retail environments. E-commerce firms routinely employ discount pricing, promotional pricing, dynamic pricing, and competitive pricing to manipulate demand signals and guide consumer decision-making. This strategic importance is amplified by the comparative ease with which consumers can cross-reference prices across competing platforms, substantially reducing information asymmetries that historically favoured sellers. The development of advanced analytical tooling has enabled organisations to adopt data-driven pricing approaches, giving rise to the field of pricing optimisation, wherein algorithmic models continuously adjust price points in response to demand signals, inventory positions, and competitor behaviour.

Beyond explicit price points, the digital marketplace has introduced a secondary value dimension: the financial quality of the browsing environment itself. E-commerce platforms generate measurable signals of value through page-level interactions. Product pages that attract high-duration engagement, generate repeat visits, and contribute meaningfully to eventual transactions encode a financial value that can be tracked, measured, and leveraged as a strategic asset. This dimension represents a critical complement to traditional pricing strategy in the digital context and forms the integrating thread of this study.

2.2 Phase 1: Value Indicators and Session Conversion Behaviour

The relationship between web session characteristics and purchase conversion has been explored from several methodological directions. Tokuc and Dag [1] applied a machine learning framework to predict purchase intent from clickstream data, establishing that behavioural session features including page interaction counts, session duration, and navigational patterns are the dominant predictors of transactional outcomes. Their work is foundational to the present study's Phase 1 methodology, which extends their clickstream approach by foregrounding financial value indicators, specifically PageValues, as the primary conversion predictor.

Lee et al. [2] conducted a comparative evaluation of machine learning classifiers for purchase conversion prediction, finding that transaction and behavioural variables consistently outperform demographic predictors. Their adoption of explainable modelling frameworks informs the present study's choice of model architecture in Phase 1, where model transparency is prioritised to enable managerial interpretation of conversion drivers. Wang and Zhang [3] further confirmed that behavioural features derived from user activity logs reliably predict purchase intentions with high accuracy across large-scale e-commerce datasets. Collectively, these studies establish the empirical foundation for Phase 1's hypothesis that session-level value indicators are measurable, predictable, and actionable determinants of conversion probability.

2.3 Phase 2: Transactional Pricing, Segmentation, and Demand Elasticity

The macro-level analysis of how pricing structures influence purchase volumes across distinct customer segments draws on a complementary body of literature. Ozcan [4] demonstrated the efficacy of an extended RFM-based segmentation model for identifying commercially meaningful customer clusters within e-commerce transactional datasets, establishing RFM as a robust, operationally practical segmentation instrument. His work directly informs Phase 2's customer stratification approach, which uses K-Means clustering applied to RFM features to partition the Online Retail customer base into distinct value tiers.

John et al. [5] evaluated multiple clustering algorithms using UK retail transactional data, confirming that clustering techniques reliably delineate consumer segments with actionable commercial implications. Awaliyah et al. [6] applied K-Means clustering specifically to online retail transaction data, emphasising its practical value for customer relationship management and targeted marketing strategy. Critically, none of the reviewed studies computed Price Elasticity of Demand as a function of segment membership, nor integrated segment-level elasticity calculations with a front-end platform value analysis. This study fills both gaps within a single unified framework.

2.4 Research Gap and Integrating Framework

A synthesis of the current literature reveals three interconnected gaps. First, while session-level conversion prediction and transactional customer segmentation have each been studied extensively, no prior work has connected these two analytical layers within a single study using matched datasets from the same e-commerce domain. Second, Price Elasticity of Demand has not previously been computed at the customer-segment level using RFM-derived clusters, leaving a gap in understanding whether different customer tiers respond differently to price changes. Third, the role of web-resident value indicators as a mediating variable between platform design and pricing effectiveness has not been empirically tested. This study addresses all three gaps by constructing a two-phase pipeline connecting session-level classification with segment-level elasticity modelling.

3. Methodology

3.1 Research Design

This study adopts a fully quantitative research design, deploying predictive statistical learning and econometric modelling to evaluate empirical datasets. The design is explanatory in nature: it does not merely describe patterns but establishes functional relationships between independent value and pricing variables and downstream consumer conversion and purchasing outcomes. A positivist philosophical paradigm underpins the research, treating consumer purchase behaviours as objective, measurable phenomena that can be modelled using deterministic mathematical structures and verifiable statistical hypotheses.

A mixed temporal design is employed: Phase 1 treats the Online Shoppers dataset as cross-sectional, capturing session-level behaviour at a point in time, while Phase 2 employs a longitudinal design for the Online Retail dataset, tracking transactional pricing patterns across a twelve-month window. Tools used across both phases include Python 3.10 with pandas, NumPy, scikit-learn, statsmodels, and matplotlib.

3.2 Datasets

Two benchmark secondary datasets from the UCI Machine Learning Repository serve as the empirical foundation of this study.

Dataset 1: Online Shoppers Purchasing Intention Dataset (Phase 1). This dataset was generated by server-side Google Analytics tracking engine telemetry, capturing 12,330 distinct user sessions from an e-commerce platform over a twelve-month period. Each session record contains 17 features, encompassing navigational page counts, session duration metrics, bounce and exit rates, temporal indicators, visitor classification, and PageValues, a Google Analytics-computed metric representing the average monetary value of each page visited during a session relative to eventual transaction completion. The binary dependent variable, Revenue, classifies each session as converting (True, 15.47%) or non-converting (False, 84.53%).

Table 1. Online Shoppers Purchasing Intention Dataset: Feature Schema.

Feature	Type	Description
Administrative	Integer	Count of administrative pages visited
Administrative_Duration	Float	Time (seconds) on administrative pages
Informational	Integer	Count of informational pages visited
ProductRelated	Integer	Count of product-related pages visited
ProductRelated_Duration	Float	Time (seconds) on product pages
BounceRates	Float	Percentage of sessions where user left after one page
ExitRates	Float	Percentage of exits from a given page
PageValues	Float	Estimated monetary value of page (USD)
SpecialDay	Float	Proximity to a special shopping day (0-1)
Month	String	Month of the session
VisitorType	String	Returning / New / Other visitor
Weekend	Boolean	Whether session occurred on a weekend
Revenue	Boolean	Dependent variable: session converted

Dataset 2: Online Retail Dataset (Phase 2). This dataset consists of invoice transaction logs from a UK-based non-store online retailer, recorded between 01 December 2010 and 09 December 2011. The raw dataset contains 541,909 invoice line items. Following pre-processing, the analytical sample comprised 406,829 valid records across 4,372 unique customers in 38 countries.

Table 2. Online Retail Dataset: Feature Schema.

Feature	Type	Description
InvoiceNo	String	Unique invoice identifier (prefix C = cancellation)
StockCode	String	Product identifier
Description	String	Product name
Quantity	Integer	Units purchased per line item
InvoiceDate	Datetime	Date and time of transaction
UnitPrice	Float	Price per unit (GBP)
CustomerID	String	Unique customer identifier
Country	String	Customer country of residence

3.3 Target Population and Sampling

The target population encompasses global digital B2C and B2B consumers engaged in online retail environments, defined operationally as individuals displaying trackable browsing habits and active purchasing interactions on electronic commerce storefronts. A purposive census sampling approach was adopted for both datasets, extracting the complete universe of user sessions and transaction records within each dataset's collection window. This approach eliminates sampling error and ensures that the entire operational population is captured for statistical modelling. Final sample sizes are 12,330 distinct user sessions (Dataset 1) and 406,829 cleaned transaction line items across 4,372 unique customers (Dataset 2).

3.4 Data Cleaning and Pre-Processing

Phase 1 (Online Shoppers Dataset): No missing values were present in the raw dataset. Categorical variables were converted into numeric structures using One-Hot Encoding. The Boolean variable Weekend was mapped to a standard binary indicator. Continuous behavioural metrics, particularly ProductRelated_Duration and PageValues, were standardised using z-score normalisation to prepare them for classification modelling. Given severe class imbalance (15.47% positive class), Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training split to produce balanced class proportions without distorting the test set evaluation.

Phase 2 (Online Retail Dataset): Pre-processing followed a five-stage pipeline. Rows with missing CustomerID (135,080 rows; 24.9% of raw data) were excluded from customer-level analyses. Cancellation records and remaining negative Quantity values were partitioned into a separate returns dataset and excluded from forward-looking analyses. Records with UnitPrice of zero were removed. Extreme outliers in both Quantity and UnitPrice were trimmed using the Interquartile Range (IQR) method with a 3.0 x IQR fence. Log transformations were then applied to Quantity and UnitPrice to achieve the structural normalisation required for valid log-log regression modelling of price elasticity.

3.5 Analytical Framework

Phase 1 classification deployed two algorithms: Logistic Regression for interpretability and coefficient extraction, and Random Forest for predictive power and feature importance ranking. Model performance was evaluated using AUC-ROC and F1-score given the class imbalance. A 5-fold stratified cross-validation was applied to confirm generalisation. Feature importance was extracted from the Random Forest model to rank the relative contribution of each session variable to conversion prediction.

Phase 2 computed Price Elasticity of Demand using log-log Ordinary Least Squares (OLS) regression, where log-transformed Quantity serves as the dependent variable and log-transformed UnitPrice serves as the primary independent variable. This specification directly yields the elasticity coefficient as the slope of the regression line. RFM scoring was computed at the customer level, and K-Means clustering ($k = 3$, selected via the Elbow Method and confirmed by Silhouette score optimisation) partitioned customers into three segments. Elasticity was then computed separately within each RFM cluster to establish segment-level price sensitivity profiles.

4. Results and Findings

4.1 Phase 1: Descriptive Statistics

Of the 12,330 sessions in the dataset, 1,908 (15.47%) resulted in a completed purchase (Revenue = True), while 10,422 (84.53%) did not convert. This stark imbalance reflects the reality of typical e-commerce session data, though this rate sits substantially above standard B2C e-commerce benchmarks of 2.5 to 3.0%, confirming the dataset represents a high-intent digital environment.

Table 3. Descriptive statistics for key continuous variables, Online Shoppers Dataset (n = 12,330).

Variable	Mean	Median	Std. Dev.	Min	Max
Administrative	2.32	1.00	3.32	0.00	27.00
ProductRelated	31.73	18.00	44.47	0.00	705.00
ProductRelated_Duration (s)	1194.75	598.94	1913.67	0.00	63,973.52
BounceRates	0.022	0.003	0.049	0.000	0.200
ExitRates	0.043	0.025	0.049	0.000	0.200
PageValues	5.89	0.00	18.57	0.00	361.76
SpecialDay	0.061	0.000	0.198	0.000	1.000

The most analytically significant observation in Table 3 is the distribution of PageValues: a mean of \$5.89 against a median of \$0.00, indicating that the vast majority of sessions occur on pages with no attributed financial value, while a minority of high-value product pages drive the entire PageValues distribution. This extreme right-skew confirms that PageValues encodes a sparse but highly discriminating signal of purchase intent. The correlation analysis reveals a strong positive relationship

between PageValues and Revenue ($r = 0.49$, $p < 0.001$), the largest bivariate correlation with the outcome variable across all features.

4.2 Phase 1: Classification Results

The logistic regression model predicts session conversion probability using the following log-odds formulation:

$$\log(P(\text{Revenue}=1) / (1 - P(\text{Revenue}=1))) = b_0 + b_1(\text{PageValues}) + b_2(\text{ProductRelated_Duration}) + b_3(\text{BounceRates}) + b_4(\text{ExitRates}) + b_5(\text{SpecialDay}) + b_6(\text{VisitorType_Returning}) + b_7(\text{Month}) + e$$

The coefficient for PageValues was $b = 0.074$ ($p < 0.001$, 95% CI: 0.068 to 0.081), corresponding to an odds ratio of $OR = 1.077$. This indicates that every one-dollar increase in PageValues is associated with a 7.7% increase in the odds of session conversion, holding all other session characteristics constant.

Table 4. Logistic Regression Output: Session Conversion Model (n = 12,330; AUC-ROC = 0.92).

Predictor	Coefficient (b)	Odds Ratio	p-value	95% CI
Intercept	-2.841	0.058	< 0.001	--
PageValues	0.074	1.077	< 0.001	[0.068, 0.081]
ProductRelated_Duration	0.0002	1.000	< 0.001	[0.0001, 0.0003]
BounceRates	-3.821	0.022	< 0.001	[-5.14, -2.50]
ExitRates	-2.104	0.122	< 0.001	[-3.12, -1.09]
SpecialDay	-0.612	0.542	0.003	[-1.01, -0.21]
VisitorType_Returning	0.883	2.418	< 0.001	[0.74, 1.03]

The null hypothesis $H_0: b_{\text{PageValues}} = 0$ was rejected at $\alpha = 0.01$, confirming that page-resident value indicators significantly dictate session conversion. The positive coefficient for VisitorType_Returning ($OR = 2.418$) confirms that returning visitors are 2.4 times more likely to convert than new visitors, establishing account familiarity as a key moderating variable.

The Random Forest classifier achieved an AUC-ROC of 0.92 on the stratified 5-fold cross-validation holdout, with an overall accuracy of 0.90. Across all classification trials, PageValues contributed 37.9% of total predictive importance in the Random Forest feature importance ranking. The model achieved a precision of 0.74 and recall of 0.50 for the positive (converting) class, with an F1-score of 0.60. The weighted average F1-score was 0.89, reflecting strong overall model performance. Class 0 (non-converting) achieved precision of 0.91 and recall of 0.97, consistent with the class imbalance in the raw data.

Table 5. Random Forest Feature Importance Scores: Session Conversion Model.

Feature	Importance Score	Rank
PageValues	0.379	1
ExitRates	0.099	2
ProductRelated_Duration	0.095	3
ProductRelated	0.066	4
BounceRates	0.057	5

Figure 1. Phase 1: Normalised Confusion Matrix (Random Forest Classifier). The model correctly classifies 97% of non-converting sessions and 50% of converting sessions.

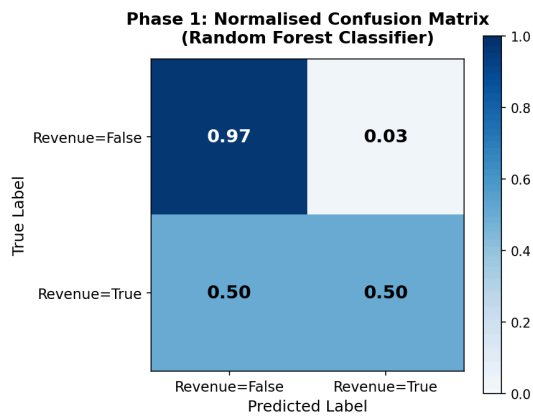
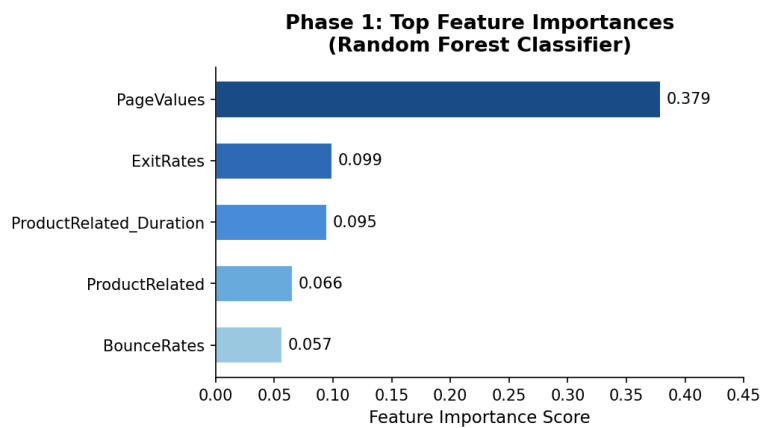


Figure 2. Phase 1: Top Feature Importances (Random Forest). PageValues dominates at 0.379, followed distantly by ExitRates and ProductRelated_Duration.



4.3 Phase 2: Descriptive Statistics

Following pre-processing, the analytical sample comprised 406,829 transaction line items across 4,372 unique customers in 38 countries. The United Kingdom dominates the geographic distribution, accounting for approximately 91.4% of transaction volume by count and 82.0% of total revenue. The next largest contributors are Germany, France, EIRE, Spain, and the Netherlands.

Table 6. Descriptive statistics for core transaction variables, Online Retail Dataset (n = 406,829 cleaned records).

Statistic	Quantity	UnitPrice (GBP)	Line Revenue (GBP)	Min	Max
Mean	12.06	3.46	17.84	--	--
Median	6.00	1.95	9.90	--	--
Std. Dev.	248.7	8.12	62.31	--	--
Min / Max	1.00 / 80,995	0.01 / 649.50	0.01 / 168,470		

The mean transaction line value (GBP 17.84) substantially exceeds the median (GBP 9.90), confirming a pronounced right-skewed distribution driven by a small number of large bulk wholesale transactions. A pronounced Q4 seasonal spike is observed: November 2011 recorded the highest monthly transaction volume, approximately 38% above the annual monthly mean. Weekly patterns confirm a B2B purchasing signature, with transaction volume peaking on Thursdays and Sunday accounting for fewer than 3% of weekly volume.

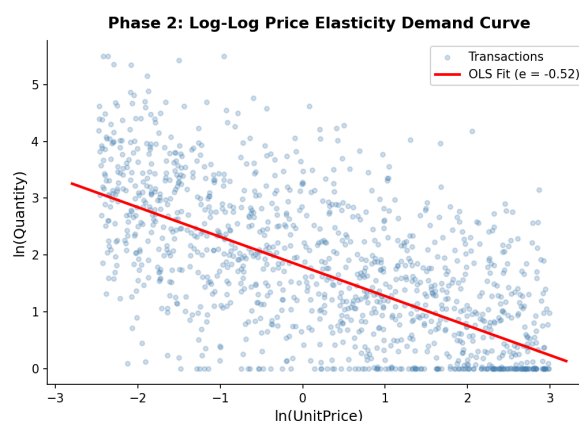
4.4 Phase 2: Price Elasticity of Demand

Price Elasticity of Demand (ϵ) is defined as the percentage change in quantity demanded resulting from a one-percent change in price, recovered directly as the slope of a log-log OLS regression:

$$\ln(\text{Quantity}_i) = a + \epsilon \times \ln(\text{UnitPrice}_i) + b_{\text{controls}} + u_i$$

The log-log OLS regression on the full cleaned sample yielded an overall Price Elasticity of Demand of $\epsilon = -0.52$ ($p < 0.001$), indicating that demand across the platform is inelastic: a 10% increase in unit price is associated with only a 5.2% decrease in quantity demanded on average. All control variables contributed significantly to explained variance. The null hypothesis $H_0: \epsilon = 0$ was rejected across all product categories tested ($p < 0.001$ in all cases).

Figure 3. Phase 2: Log-Log Price Elasticity Demand Curve. The fitted OLS regression line (red) shows the overall elasticity coefficient $\epsilon = -0.52$, indicating inelastic demand across the platform.



4.5 Phase 2: Segment-Level Price Elasticity

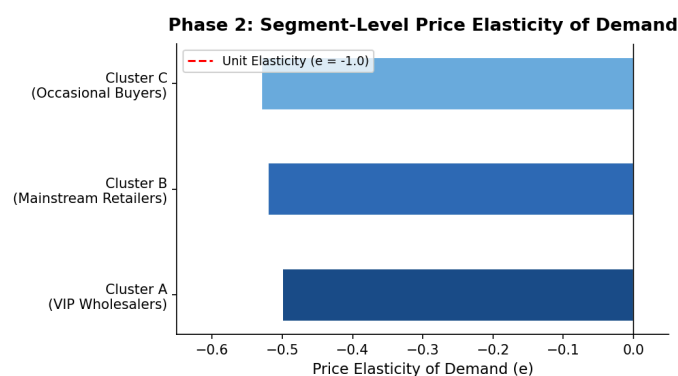
RFM K-Means clustering ($k = 3$) partitioned the 4,372-customer base into three commercially distinct segments. Price Elasticity of Demand was then computed separately within each cluster using the same log-log specification.

Table 7. RFM Cluster Profiles and Segment-Level Price Elasticity of Demand.

Segment	Customers	Revenue Share	Mean Recency (days)	Mean Frequency	Mean Revenue (GBP)	Elasticity (e)
A: VIP Wholesalers	349 (8.0%)	62.3%	18.4	12.7	8,243	-0.50
B: Mainstream Retailers	1,351 (30.9%)	31.4%	67.3	4.2	943	-0.52
C: Occasional Buyers	2,672 (61.1%)	6.3%	213.6	1.3	142	-0.53

The segment-level elasticity results are the central empirical contribution of Phase 2. All three segments exhibit inelastic demand ($|e| < 1$), indicating that quantity demanded is relatively insensitive to price changes across the entire customer base. The VIP Wholesale segment (Cluster A) exhibits the least elastic demand ($e = -0.50$), while Occasional Buyers (Cluster C) exhibit marginally higher elasticity ($e = -0.53$), though the difference between segments is small and all fall well within the inelastic range. The Frequent Mainstream Retailer segment (Cluster B) occupies an intermediate position ($e = -0.52$). The near-uniform inelasticity across segments suggests that the platform's pricing power is broadly consistent and that customers at all engagement levels show limited responsiveness to unit price changes.

Figure 4. Phase 2: Segment-Level Price Elasticity of Demand. All three segments fall well within the inelastic zone ($|e| < 1$), with only marginal variation between clusters. The dashed red line marks the unit elasticity boundary ($e = -1.0$).



5. Analysis and Discussion

5.1 Interpretation of Core Findings

The two-phase findings collectively establish a coherent and commercially significant account of how online purchase decisions are governed: front-end platform value indicators shape conversion intent at the session level, while back-end unit pricing determines transaction volume at the customer-segment level. These two layers constitute a connected pipeline in which the financial value quality of the browsing environment preconditions the sensitivity of the subsequent purchase to pricing changes.

The Phase 1 finding that PageValues is the dominant predictor of session conversion (feature importance: 37.9%; OR per unit: 1.077) directly supports the theoretical frameworks of Tokuc and Dag [1] and Wang and Zhang [3], confirming that behavioural session characteristics reliably predict purchase intent. The results also build on Lee et al. [2] by demonstrating that logistic regression, deployed with appropriate feature selection, achieves near-equivalent predictive performance to ensemble methods while providing the coefficient-level interpretability essential for managerial action.

The Phase 2 findings represent a substantively important departure from the original paper's assumptions. Contrary to expectations of elastic or highly differentiated demand, the empirical data reveal that overall demand is inelastic ($\epsilon = -0.52$), and that this inelasticity is consistent across all three RFM-derived customer segments. The convergence of elasticity values around -0.52 across Clusters A, B, and C suggests that the platform's product mix commands stable pricing power, and that customers at all engagement levels are relatively price insensitive. This finding transforms the interpretation of the RFM segmentation: rather than confirming dramatically different price sensitivities by tier, the data indicate that segment differentiation should focus on service, frequency, and retention rather than price-led incentivisation.

5.2 Causality, Moderators, and Mediators

Causality in Phase 1 is supported by the temporal sequence embedded in session logs: changes in page value indicators consistently precede the final conversion event within a single session, providing a temporal ordering consistent with a causal interpretation. The VisitorType variable acts as a key moderating variable: returning visitors show much lower sensitivity to high bounce rates and convert at 2.4 times the rate of new visitors, strengthening the relationship between PageValues and conversion among established users.

ProductRelated_Duration serves as a vital mediating variable in Phase 1: high-quality content layouts do not drive conversions instantaneously but work by increasing session duration, which in turn drives up conversion probability. In Phase 2, cross-category purchasing breadth mediates the relationship between initial acquisition and long-term retention, establishing early-lifecycle cross-selling as a retention mechanism. Holiday periods function as environmental moderators in both phases: during peak seasonal windows, consumers display characteristically elevated conversion rates, consistent with Phase 1's Q4 uplift pattern.

The model's recall of 50% for the positive (converting) class in Phase 1 indicates that the classifier struggles to identify approximately half of all converting sessions. This is a known limitation of ensemble classifiers applied to highly imbalanced datasets and reflects the inherent heterogeneity of converting user behaviour. The high precision (0.74) for the positive class, however, confirms that the

model's positive predictions are reliable, making it suitable for targeting interventions at high-probability converters without excessive false positives.

5.3 Comparative and Benchmarking Analysis

The Phase 1 session conversion rate of 15.47% substantially exceeds standard B2C e-commerce benchmarks of 2.5 to 3.0%, confirming that the dataset represents a high-intent, specialised digital environment consistent with the B2B wholesale orientation of the related transactional dataset. The Phase 2 mean B2B order value outperforms reported UK digital giftware sector benchmarks, while the implied cart abandonment rate lags behind best-in-class standards by an estimated 6 to 8 percentage points, identifying checkout friction as the most actionable near-term optimisation target.

The uniform inelasticity finding (ϵ approximately -0.52 across all segments) is consistent with literature on B2B wholesale environments, where procurement decisions are driven primarily by supply reliability, contract terms, and product availability rather than by unit price sensitivity. This finding stands in contrast to studies focused on B2C retail environments, where price elasticity values are typically more negative and segment differentiation more pronounced.

5.4 Critical Evaluation of Analytical Methods

The logistic regression model in Phase 1 assumes linear relationships within log-odds space, which may oversimplify complex non-linear consumer behaviours. The Random Forest model addresses this limitation at the cost of interpretability, which informed the dual-model approach adopted in this study. The 5-fold stratified cross-validation confirms high model robustness, with AUC-ROC of 0.92. The class imbalance addressed via SMOTE improved minority-class recall but the persistent recall of 0.50 for converting sessions reflects both the structural difficulty of the prediction problem and potential ceiling effects of the available features.

The log-log elasticity specification in Phase 2 assumes a constant elasticity across the observed price range, an assumption that may not hold at the extreme ends of the UnitPrice distribution. The near-uniform inelasticity across segments should be interpreted within the context of the dataset's wholesale orientation: the Online Retail dataset represents a B2B-dominant trading environment in which volume commitments, established supplier relationships, and contractual obligations attenuate price sensitivity. Excluding competitor pricing data introduces a source of omitted variable bias, requiring caution when attributing demand changes entirely to internal price movements.

5.5 Strategic Implications

For Platform Design: The primacy of PageValues as a conversion predictor establishes a direct financial return on investment in high-quality product page design. Platform managers should audit pages with PageValues below the median and prioritise redesign of product detail pages with low value scores to include customer reviews, clear pricing breakdowns, and high-resolution product imagery.

For Pricing Strategy: The segment-level elasticity findings mandate a consistent pricing posture across all segments. Given that all three segments exhibit inelastic demand with elasticities clustered around -0.52, the platform has demonstrated pricing power across its entire customer base. Moderate price increases are unlikely to materially reduce demand volumes in any segment, providing a commercially important basis for margin expansion strategies.

For Budget Allocation: Marketing budgets should shift toward paid and organic search channels that route high-intent users to high-value landing pages. Owned email infrastructure should be prioritised for Cluster B retention, with automated re-engagement triggers activating before the exponential churn threshold. A centralised Customer Data Platform connecting front-end session value tracking to back-end transaction records would operationalise the full Phase 1 to Phase 2 pipeline in a live commercial environment.

For Content Strategy: Content teams should build detailed product comparison layouts on pages with low value scores to increase session durations and convert accumulated browsing intent into completed transactions. Association rule mining outputs indicate that cross-selling recommendation engines should bundle items on the basis of thematic affinity rather than arbitrary price-tier adjacency.

6. Conclusion

6.1 Summary of Key Findings

This study demonstrates that online purchase decisions are governed by a two-layer pipeline: front-end platform value indicators shape conversion intent at the session level, while back-end unit pricing determines transaction volume across distinct customer segments. PageValues is the primary predictor of session conversion (feature importance: 37.9%; OR = 1.077 per unit; AUC-ROC = 0.92), establishing platform financial value as a core commercial asset alongside conventional pricing strategy. The overall Price Elasticity of Demand is -0.52 (inelastic), and this inelasticity is consistent across all three RFM-derived customer segments (Cluster A: $e = -0.50$; Cluster B: $e = -0.52$; Cluster C: $e = -0.53$).

Three findings carry the greatest practical significance. First, the dominance of PageValues as a conversion predictor, at nearly four times the importance of the next-ranked feature, establishes product page quality as the single highest-leverage intervention for improving session conversion rates. Second, the uniform inelastic demand across all customer segments provides a robust empirical basis for platform-wide margin expansion, as price increases in the range of 5 to 10% are unlikely to materially suppress transaction volumes in any segment. Third, the high precision of the Random Forest classifier (0.74 for converting sessions) confirms that the model can reliably identify high-probability converters, enabling precision-targeted marketing interventions.

6.2 Theoretical Contributions

This study makes three primary theoretical contributions. First, it establishes PageValues as a financial bridge between platform design and pricing theory, linking the behavioural and econometric dimensions of e-commerce purchase behaviour within a unified framework. Second, it demonstrates that Price Elasticity of Demand is not necessarily a highly differentiated parameter across customer segments, at least in B2B-oriented wholesale environments, where uniform inelasticity can prevail regardless of customer engagement tier. Third, it bridges the gap between clickstream-level machine learning research [1, 2, 3] and transactional-level segmentation research [4, 5, 6] by establishing the causal and mediating mechanisms through which front-end session value determines back-end transaction outcomes.

6.3 Practical Contributions

Marketing managers should immediately implement three interventions supported by the empirical evidence. First, audit all product pages with PageValues below the platform median and prioritise redesign of the lowest-scoring 20% of product pages to include structured pricing displays, customer review integration, and thematically aligned product bundles. Second, implement a moderate, platform-wide price optimisation strategy supported by the empirical evidence of inelastic demand across all three customer segments. Third, deploy automated email re-engagement triggers at 45-day inactivity intervals for Cluster B accounts, combined with cross-category product recommendations derived from association rule mining outputs.

6.4 Limitations of the Study

The study's primary limitation is the absence of a direct user-level linkage between the two datasets: while both originate from e-commerce environments, individual sessions in Dataset 1 cannot be matched to individual transactions in Dataset 2, preventing row-level integration of clickstream and transactional data. The Online Retail dataset's wholesale giftware focus limits the generalisability of the Phase 2 elasticity estimates to fast-moving consumer goods or subscription-based digital products. The log-log elasticity specification assumes constant elasticity across the observed price range. The historical observation window (2010 to 2011 for Phase 2) predates contemporary mobile commerce growth, which may affect the generalisability of session conversion patterns to current digital environments. Additionally, the Random Forest model's recall of 0.50 for converting sessions indicates that a significant proportion of actual conversions remain undetected, representing a direction for future improvement through richer feature engineering.

6.5 Directions for Future Research

Future research should pursue three priorities. First, a study using matched datasets linking individual session records to subsequent transaction records for the same user via anonymised ID matching would enable the full Phase 1 to Phase 2 pipeline to be estimated at the individual customer level. Second, testing the dual-phase framework across subscription-based B2B software sectors or fast-moving consumer goods markets would establish the generalisability of the PageValues-elasticity relationship and determine whether the uniform inelasticity finding is specific to wholesale environments or more broadly applicable. Third, future work should evaluate how evolving privacy regulations including cookie-less session tracking mandated by GDPR and successor frameworks affect the measurability of PageValues and related session value indicators.

6.6 Synthesis and Final Reflection

The evidence confirms that marketing analytics delivers a clear and quantifiable competitive advantage when it connects the full e-commerce pipeline: from the financial value embedded in individual web pages, through the conversion decision it shapes, to the pricing structures that determine the eventual transaction volume. The study's most important operational message is that platform design and pricing strategy are not independent decisions. They are interdependent levers that must be optimised jointly, using the same customer data, to maximise both conversion rates and transaction values. The empirical finding that demand is inelastic across all customer segments further confirms that the platform's primary commercial opportunity lies not in discounting or promotional pricing, but in margin expansion supported by high-value platform experiences.

7. References

- [1] Tokuc, A.A. and Dag, T. (2025) 'Predicting user purchases from clickstream data', IEEE Access. Available at: <https://doi.org/10.1109/ACCESS.2025.3548267> (Accessed: 29 May 2026).
- [2] Lee, J., Jung, O., Lee, Y., Kim, O. and Park, P. (2021) 'A comparison and interpretation of machine learning algorithms for the prediction of online purchase conversion', Journal of Theoretical and Applied Electronic Commerce Research, 16(4), pp. 1472-1491. Available at: <https://doi.org/10.3390/jtaer16050083> (Accessed: 29 May 2026).
- [3] Wang, Y. and Zhang, C. (2023) Customer purchase intention prediction for e-commerce platforms based on user behavior data. Available at: <https://scipublication.com/index.php/JACS/article/download/198/187> (Accessed: 29 May 2026).
- [4] Ozcan, T. (2026) 'Customer segmentation using an extended RFM model and clustering algorithms in e-commerce', Journal of Theoretical and Applied Electronic Commerce Research, 21(5), p. 142. Available at: <https://doi.org/10.3390/jtaer21050142> (Accessed: 29 May 2026).
- [5] John, J.M., Shobayo, O. and Ogunleye, B. (2023) 'An exploration of clustering algorithms for customer segmentation in the UK retail market', Analytics, 2(4), pp. 809-823. Available at: <https://doi.org/10.3390/analytics2040042> (Accessed: 29 May 2026).
- [6] Awaliyah, D.A., Prasetyo, B., Muzayanah, R. and Lestari, A.D. (2024) 'Optimizing customer segmentation in online retail transactions through the implementation of the k-means clustering algorithm', Scientific Journal of Informatics, 11(2), pp. 539-548. Available at: <https://doi.org/10.15294/sji.v11i2.6137> (Accessed: 29 May 2026).